7.8 Organic Synthesis (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.8 Organic Synthesis (A Level Only)
Difficulty	Medium

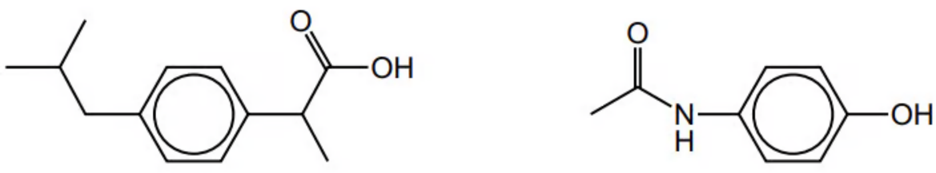
Time allowed: 80

Score: /65

Percentage: /100

Question 1a

Ibuprofen and paracetamol are pain-relief drugs.



ibuprofen paracetamol

Fig. 8.1

Ibuprofen and paracetamol both contain the aryl (benzene) functional group.
Name the other functional groups present in each molecule.

ibuprofen ....

paracetamol ....

[2 marks]

Question 1b

Ibuprofen contains a chiral centre and has two enantiomers.

i)
State one similarity and one difference in the physical or chemical properties between the two enantiomers.

similarity.......................................

difference....................................

[1]

ii)
Explain what is meant by racemic mixture.

[1]

[2 marks]

Question 1c

Paracetamol reacts separately with the two reagents shown in the table.

Complete Table 8.1 by:
• drawing the structures of the organic products formed,
• stating the types of reaction.

Table 8.1

reagent	organic product structure	type of reaction
$LiAlH_4$		
an excess of $Br_2(aq)$		

[3 marks]

Question 1d

One of the steps in the manufacture of ibuprofen is shown in Fig. 8.2.

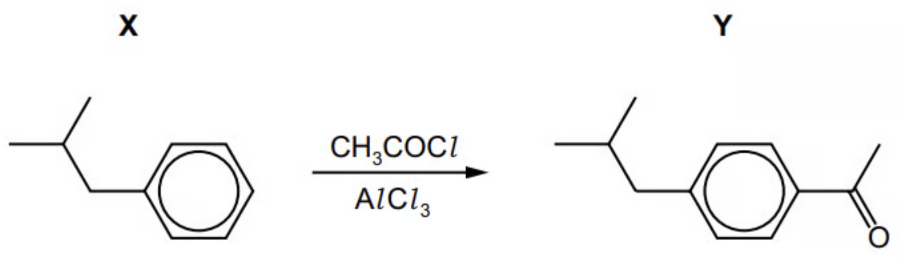


Fig. 8.2

i)

Write an equation to show how $AlCl_3$ generates the electrophile for the conversion of **X** into **Y.**

[1]

ii)

Draw the mechanism for the conversion of **X** into **Y**. Include all necessary curly arrows and charges.

[3]

iii)

Write an equation to show how the $AlCl_3$ is regenerated.

[1]

[5 marks]

Question 2a

Cuminyl alcohol can be synthesised from benzene by the following route shown in Fig. 2.1.

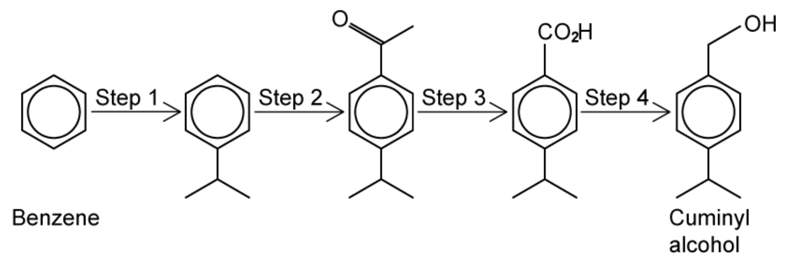


Fig. 2.1

Suggest reagents and conditions for steps 1–4.

step 1 ....

step 2 ....

step 3 ....

step 4 ....

[4 marks]

Question 2b

Name the mechanism of step 2 and state the type of reaction in step 4.

Mechanism of step 2 ....

Type of reaction in step 4 ....

[2 marks]

Question 2c

Draw the reaction mechanism for step 2.

[4 marks]

Question 2d

Deduce the number of peaks that would be present in the ^{13}C NMR spectrum of cuminyl alcohol.

[1 mark]

Question 3a

Outline how ethanal can be synthesised from ethane in three steps.

$$\text{ethane} \xrightarrow{\text{step 1}} \text{chloroethane} \xrightarrow{\text{step 2}} \text{ethanol} \xrightarrow{\text{step 3}} \text{ethanal}$$

State the reaction conditions and reagents and name the type of reaction taking place.

i)
Step 1

Reagents ....

Conditions ....

Reaction type ....

[3]

ii)
Step 2

Reagents ....

Conditions ....

Reaction type ....

[3]

iii)
Step 3

Reagents ....

Conditions ....

Reaction type ....

[3]

[9 marks]

Question 3b

The following reaction pathway in Fig. 3.1 used to produce compounds **A** and **B**, which when reacted together, form a branched ester molecule, compound **C**.

Suggest suitable reagents and conditions for the synthesis of compound **A** via step 1 and give the name for this type of reaction.

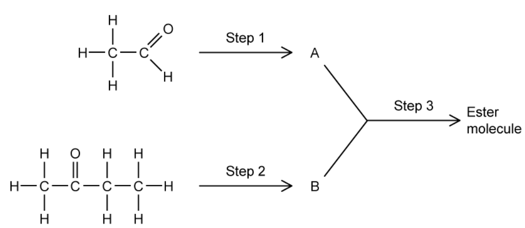


Fig. 3.1

[3 marks]

Question 3c

The ketone in part **(a)**, must be converted to compound **B** to produce the ester.

i)
Name the molecule that is produced from step 2

[1]

ii)
Name of the type of reaction that is involved in step 2 and suggest suitable reagents and conditions for this step.

[2]

[3 marks]

Question 3d

Draw the skeletal formula of the ester formed from the reaction scheme.

[1 mark]

Question 4a

Noradrenaline shown in Fig. 4.1 is a hormone and neurotransmitter, which is released during stress to stimulate the heart and increase blood pressure.

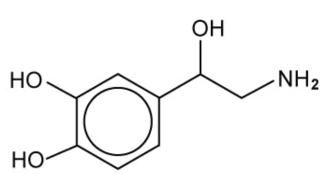


Fig. 4.1

State the names of three functional groups in the noradrenaline molecule.

[3 marks]

Question 4b

Consider the following two step synthesis of noradrenaline from dihydroxybenzaldehyde in Fig. 4.2.

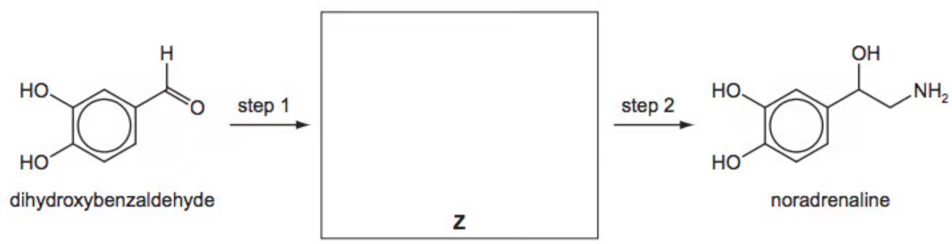


Fig. 4.2

Draw the structure of the intermediate Z in the box.

Suggest reagents for steps 1 and 2.

step 1 ....

step 2 ....

[3 marks]

Question 4c

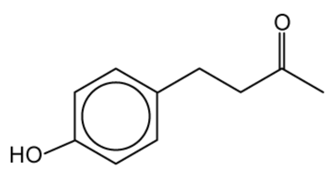
Dihydroxybenzaldehyde reacts with Br_2 (aq).

- Describe what you would see during this reaction
- Draw the structure of the product

[2 marks]

Question 5a

Compound **G** is a naturally occurring aromatic compound that is present in raspberries.



Compound **G**

Identify the functional groups present in compound **G**.

[2 marks]

Question 5b

Complete Table 5.1 with information about the reactions of the three stated reagents with compound **G**.

Table 5.1

Reagent	Observation	Structure of organic product	Type of reaction
Na (s)			
Br_2 (aq)			
I_2 / NaOH (aq)			

[9 marks]

Question 5c

The dye H can be made from compound **G** by the route shown below in Fig. 5.1.

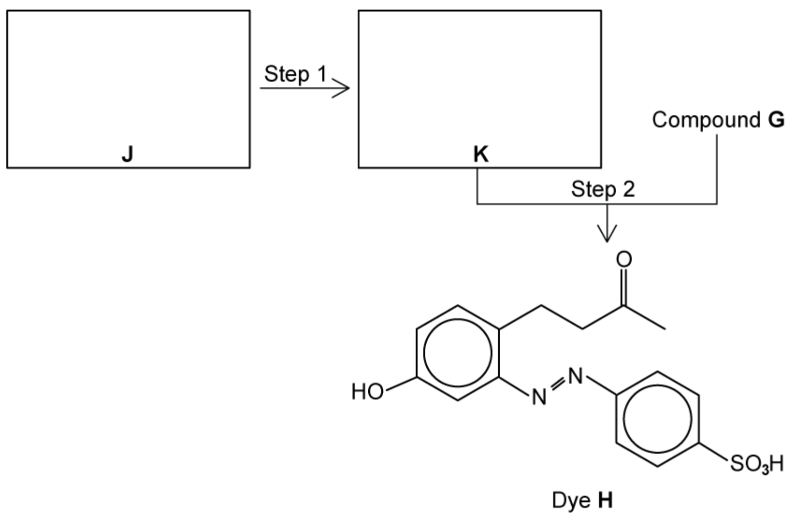


Fig. 5.1

i)
Draw the structures of the amine **J** and the intermediate **K** in the boxes above.

[2]

ii)
Suggest reagents and conditions for

step 1 ....

step 2 ....

[3]

[5 marks]

Question 5d

Explain why dye **H** is very stable.

[2 marks]

Model Answers: Medium

1a

a) The other functional groups present in each molecule are:

- Ibuprofen: carboxylic acid; [1 mark]
- Paracetamol: phenol
  AND
  Amide; [1 mark]

[Total: 2 marks]

- An OH group attached to an arene ring changes its chemistry completely
  - This is why it is termed phenol rather than alcohol
- The paracetamol contains a secondary amide, although you don't need to classification to score the mark
  - A primary amide would have two hydrogens on the N

1b

b)

i) One similarity and one difference in the physical or chemical properties between the two enantiomers are:

- Similarity: Enantiomers have identical physical and chemical properties
  AND
  Difference: The ability to rotate plane polarised light / potential biological activity; [1 mark]

ii) The meaning of racemic mixture:

- A mixture containing equal amounts of each enantiomer; [1 mark]

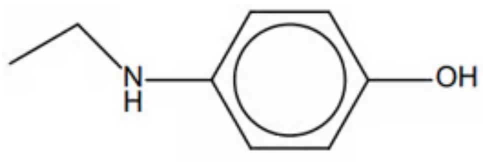
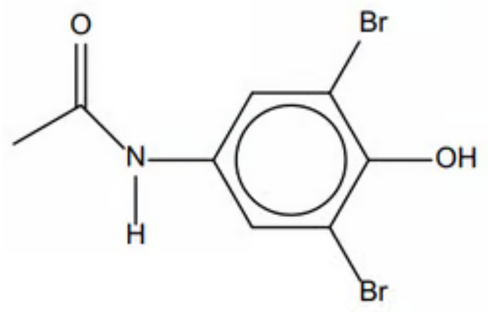
[Total: 2 marks]

- Enantiomers are mirror-image molecules containing a chiral carbon, that is, a

carbon with four different groups attached
- Another term for racemic mixture is a racemate

1c

c) The completed table:

reagent	organic product structure	type of reaction
$LiAlH_4$	 [1 mark]	**reduction**
an excess of $Br_2(aq)$	 [1 mark]	**electrophilic substitution**

- Both correct reaction types; [1 mark]

[Total: 3 marks]

- You should know that lithium aluminium hydride is a reducing agent, so look for functional groups that can be reduced.
  - Amides can be reduced to amines
  - The products of a non-substituted amide are: a primary amine and water
  - The products of a substituted amide are: a secondary amine and water
  - Paracetamol contains a secondary amide, so a secondary amine and water are formed
- Due to the phenolic OH group, bromination would be directed to the 2, 4 and 6 positions
  - There is already a functional group attached in the 6 position so bromine will add to the 2 and 4 positions

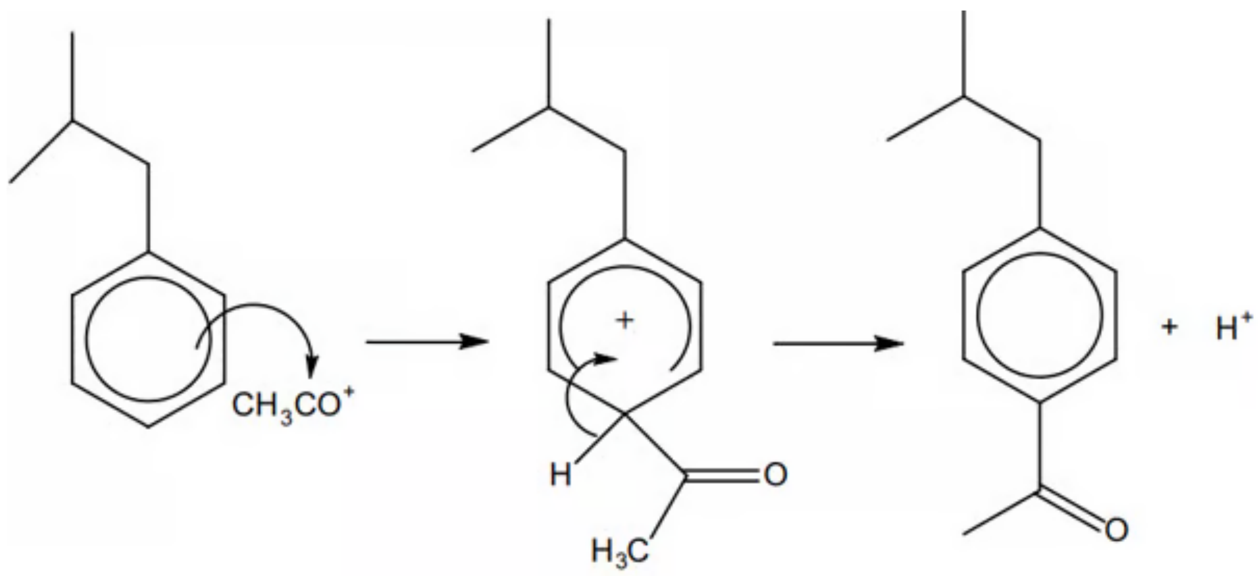
1d

d)

i) An equation to show how $AlCl_3$ generates the electrophile for the conversion of **X** into **Y** is:

- $CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$; [1 mark]

ii) The mechanism for the conversion of **X** into **Y**. Include all necessary curly arrows and charges:



- Curly arrow from the ring system to the CH_3CO^+ (arrow to the positively charged carbon atom); [1 mark]
- Correct intermediate structure; [1 mark]
- The curly arrow from the C–H bond into the ring
  AND
  Loss of H^+; [1 mark]

iii) An equation to show how the $AlCl_3$ is regenerated is:

- $H^+ + AlCl_4^- \rightarrow HCl + AlCl_3$; [1 mark]

[Total: 5 marks]

- You need to know how $AlCl_3$ acts as a catalyst to form the positive ion required for the reaction, including the equations to form the ion and reform the $AlCl_3$ catalyst
- In part ii)
  - The arrows must start from inside the circle and form the middle of the C-H

bond
  - The intermediate needs to have the positive charge to be correct
  - The bond is formed between the carbon of the ring and the carbonyl carbon of the CH_3CO^+ so you must show the arrow pointing to the carbon, not the oxygen
  - Mechanisms are a good way to pick up quite a few marks so it is worth practising them
  - The origin and destination of curly arrows must be clear and unambiguous or you may lose the marks

2a

a) The reagents for each step are:

- Step 1: heat with $AlCl_3$ + $(CH_3)_2CHCl$ or $CH_3CH{=}CH_2$; [1 mark]
- Step 2: heat with $AlCl_3$ + CH_3COCl; [1 mark]
- Step 3: NaOH + I_2 (or Cl_2) (then H^+); [1 mark]
- Step 4: $LiAlH_4$ (in dry ether); [1 mark]

[Total: 4 marks]

- Step 1 involves electrophilic substitution of a benzene ring
  - In order to form the electrophile a halogen carrier which can be $AlCl_3$ if the halogenoalkane contains a chlorine atom
  - $FeBr_3$ is also acceptable if the halogenoalkane contains a Br atom
  - $AlCl_3 + (CH_3)_2CHCl \rightarrow AlCl_4^- + (CH_3)_2CH^+$
- Step 2 involves the same type of mechanism, though a CH_3CO group is being added to the benzene ring this time
  - Therefore an acyl chloride and a halogen carrier is required
  - $AlCl_3 + CH_3COCl \rightarrow AlCl_4^- + CH_3CO^+$
- Step 3 involves the formation of a carboxylic acid from a carbonyl group (a ketone in this case)
  - Therefore the iodoform reaction is required
  - $R\text{-}CO\text{-}CH_3 + 3I_2 + 4OH^- \rightarrow CH_3I + RCOO^- + 3I^- + 3H_2O$
  - The $RCOO^-$ reacts with H^+ to form the RCOOH
- Step 4
  - The final step involves a reaction with $LiAlH_4$ to form a primary alcohol from a carboxylic acid

2b

b) The mechanism for step 2 and type of reaction for step 4 are:

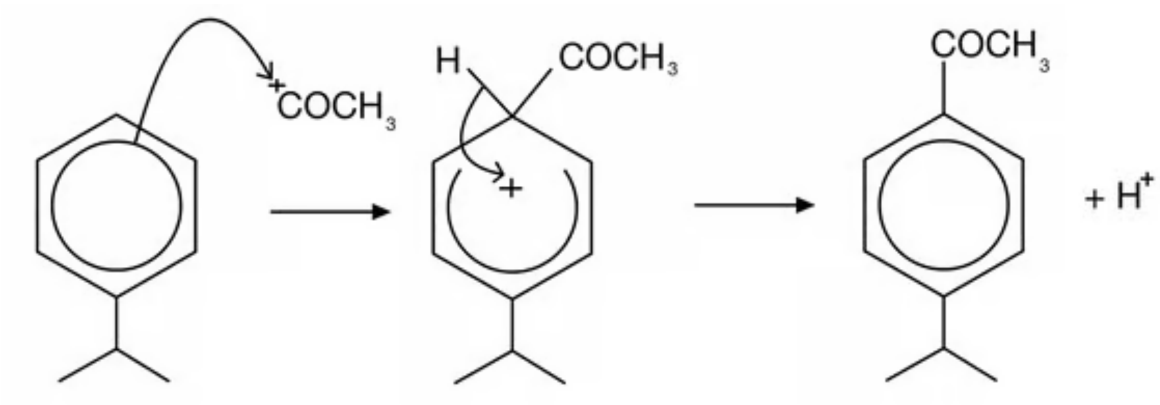
- Mechanism of step 2: Electrophilic (aromatic) substitution; [1 mark]
- Type of reaction in step 4: Reduction; [1 mark]

[Total: 2 marks]

- Arenes will undergo electrophilic substitution, whereby a hydrogen atom in the benzene ring is substituted for an attacking electrophile
  - Make sure you know the mechanism for this reaction
- The formation of a primary alcohol from a carboxylic acid involves reduction

2c

c) The reaction mechanism is:



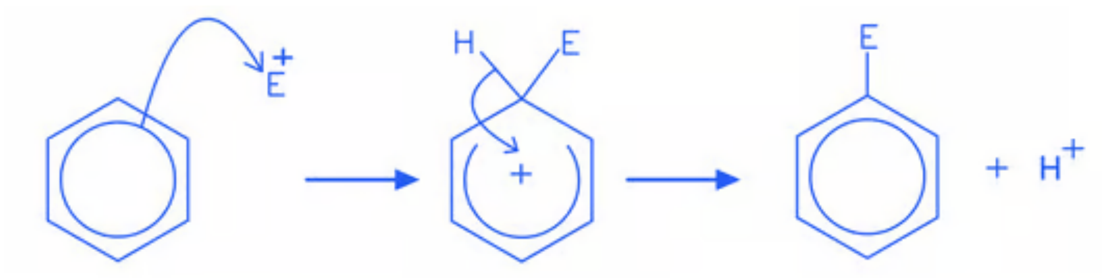
- Curly arrow from the ring within benzene to the $^{+}COCH_3$; [1 mark]
- Correct intermediate structure with horseshoe covering more than half of the ring but not extending past carbons 2 and 6; [1 mark]
- Curly arrow from the C-H bond back to the + within the ring
  AND
  Correct benzene product
  AND
  H^+ shown; [1 mark]

[Total: 3 marks]

- Benzene mechanisms are electrophilic substitution reactions and the common

mistakes are to:
- Not draw the hydrogen that is being substituted
- Draw the hydrogen being substituted in the wrong position
- Forget to put the + in the correct place on the intermediate, especially using the Kekule structure for the mechanism
- Draw the ring inside the intermediate structure incorrectly, if using the ring structure for the mechanism

- The general mechanism for electrophilic substitution is as follows:



2d

d) The number of ^{13}C peaks are:

- 7 peaks; [1 mark]

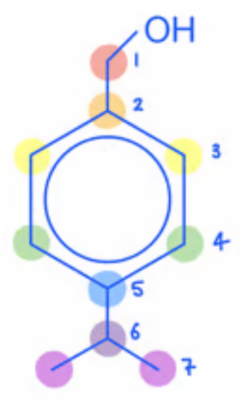
[Total: 1 mark]

d) The number of ^{13}C peaks are:

- 7 peaks; [1 mark]

[Total: 1 mark]

- **Tip:** It helps to look for symmetry within the molecule when identifying ^{13}C peaks as this identifies equivalent carbon environments quicker
- The 7 peaks in the ^{13}C spectrum for cuminyl alcohol are:



3a

a) The synthesis of ethanal from ethane is:

i) Step 1: Ethane to chloroethane

- Reagent: chlorine / Cl_2; [1 mark]
- Conditions: UV light / high temps; [1 mark]
- Reaction type: (free radical) substitution / halogenation ; [1 mark]

ii) Step 2: Chloroethane to ethanol

- Reagent: aqueous sodium hydroxide / NaOH (aq)
  OR
  Aqueous potassium hydroxide / KOH (aq); [1 mark]
- Conditions: hot / heat under reflux; [1 mark]
- Reaction type: (nucleophilic) substitution / hydrolysis ; [1 mark]

iii) Step 3: Ethanol to ethanal

- Reagent: acidified potassium dichromate(VI)
  OR

Acidified potassium dichromate(VI)
OR
Acidified potassium manganate(VII)); [1 mark]

- Conditions: distil / distillation; [1 mark]
- Reaction type: oxidation; [1 mark]

[Total: 9 marks]

- The conversion of an alkane to a haloalkane would be very impractical industrially
  - This is because it is hard to control the extent of halogenation and to prevent multiple products from forming
  - However, it is included to test the application of your knowledge of organic reactions to solve synthesis problems
- The type of mechanisms are shown in brackets, but they are not essential for the mark
- The dichromate ion, $Cr_2O_7^{2-}$, is the key oxidising agent, so it is possible to gain the mark from sodium dichromate as well as potassium dichromate

3b

a) Suitable reagents for the synthesis of compound **A** via step 1:

- $K_2Cr_2O_7$ / potassium dichromate(VI)
  OR
  $KMnO_4$ / potassium permanganate(VII); [1 mark]
- H_2SO_4 / acidified / with acid; [1 mark]
- Oxidation; [1 mark]

[Total: 3 marks]

- An ester is commonly made from a carboxylic acid and alcohol
- A carboxylic acid cannot be produced from the second molecule which is a ketone, so the first step producing **A**, must be the oxidation of an aldehyde to carboxylic acid
- **Careful:** The question asks for names of reagents
  - You could lose marks for writing ions, e.g. $Cr_2O_7^{2-}$, as these are not reagents - a reagent is something that you can find on a laboratory shelf! Sodium dichromate(VI) is, however, an acceptable alternative

3c

c)

i) The name of the molecule that is produced from step 2 is:

- Butan-2-ol; [1 mark]

ii) Give the name of the type of reaction that is involved in step 2 and suggest a suitable reagent for the process

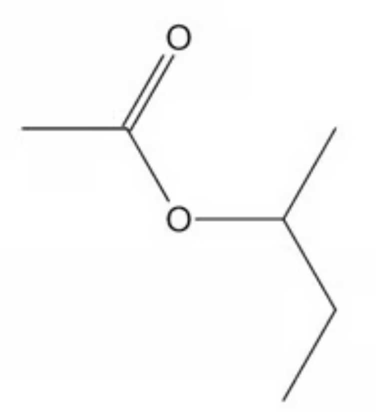
- Reduction; [1 mark]
- $LiAlH_4$ / lithium aluminium hydride / lithium tetrahydridoaluminate
  OR
  $NaBH_4$ / sodium borohydride / sodium tetrahydridoborate; [1 mark]

[Total: 3 marks]

- The reduction of a ketone produces a secondary alcohol
  - This secondary alcohol is required to produce the branched chain ester
- Lithium tetrahydridoaluminate, $LiAlH_4$, is much more reactive than sodium tetrahydridoborate, $NaBH_4$, so is more dangerous to use in the laboratory
- For the purposes of this question, both reagents are suitable answers

3d

d) The skeletal formula of the ester formed from the reaction scheme is:

- 

; [1 mark]

[Total: 1 mark]

- **Remember:** When drawing esters, the acid provides the COO portion of the ester link while the alcohol attaches to the single bonded oxygen with the elimination of a molecule of water

4a

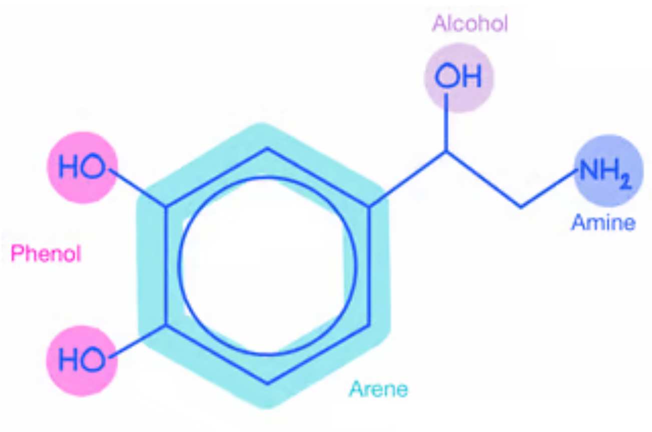
a) The three functional groups

Any **three** from the following

- Phenol; [1 mark]
- Primary alcohol; [1 mark]
- Amine; [1 mark]
- Arene / aryl / benzene; [1 mark]

[Total: 3 marks]

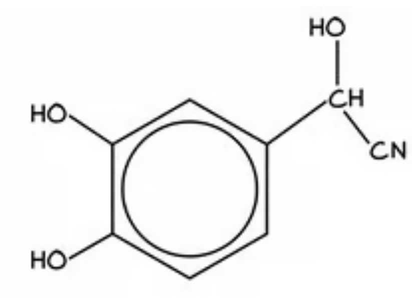
- The groups in noradrenaline are:



4b

b) Compound **Z** and the reagents for steps 1 and 2 are:

- Compound **Z** is:



; [1 mark]

- Step 1: HCN and NaCN
  OR
  HCN and base; [1 mark]

- Step 2: H_2 and Ni
  OR
  $LiAlH_4$ in dry ether
  OR
  Na and ethanol; [1 mark]

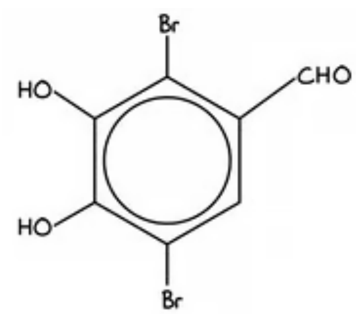
[Total: 3 marks]

- To convert the functional group from dihydroxybenzaldehyde to noradrenaline the carbon chain needs to extend
- Therefore, a nitrile group CN needs to be added to the carbonyl carbon
- Then, reduction takes place to form the amine group

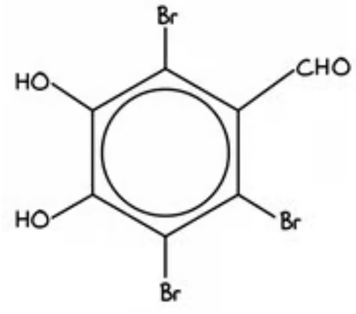
4c

c) The observation and product of this reaction are:

- Bromine decolourises
  OR
  Bromine changes from orange to colourless
  OR
  A white precipitate is formed; [1 mark]
- The structure is:



OR



; [1 mark]

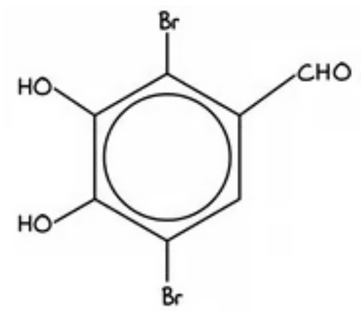
[Total: 2 marks]

- When drawing the structure you will gain a mark for including either two or three bromine atoms
- Phenols react readily with bromine and this reaction occurs at room temperature
- Dihydroxybenzaldehyde decolourises the orange bromine solution to form a white precipitate via electrophilic substitution
  - More than one hydrogen atom on the ring can be substituted as this

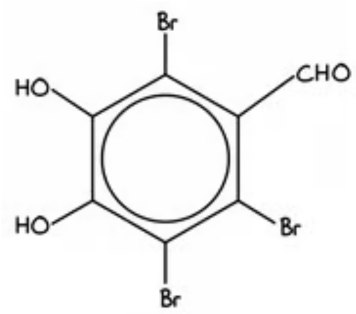
5a

a) The functional groups present in compound **G** are:

- Phenol; [1 mark]



OR



; [1 mark]

[Total: 2 marks]

- When drawing the structure you will gain a mark for including either two or three bromine atoms
- Phenols react readily with bromine and this reaction occurs at room temperature
- Dihydroxybenzaldehyde decolourises the orange bromine solution to form a white precipitate via electrophilic substitution
  - More than one hydrogen atom on the ring can be substituted as this

5a

a) The functional groups present in compound **G** are:

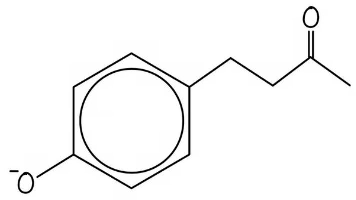
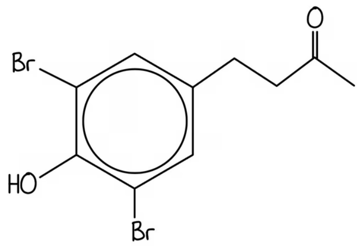
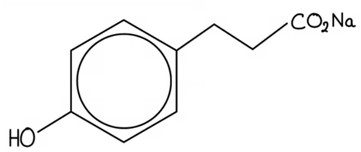
- Phenol; [1 mark]
- Ketone; [1 mark]

[Total: 2 marks]

- When an -OH group is bonded to a benzene ring it is a phenol group
- The C=O group in the carbon chain is a ketone group

5b

b) The completed table is:

Reagent	Observation	Structure of organic product	Type of reaction
Na (s)	Effervescence / bubbles / fizzing; [1 mark]	 ; [1 mark]	Redox; [1 mark]
Br_2 (aq)	Decolourises or white ppt.; [1 mark]	 ; [1 mark]	Electrophilic substitution; [1 mark]
I_2 / NaOH (aq)	Yellow ppt. ; [1 mark]	 ; [1 mark]	Oxidation; [1 mark]

[Total: 9 marks]

- Na (s) will react with the -OH group to form an O^-
  - The oxygen-hydrogen bond in the hydroxy group breaks
- Aqueous bromine reacts with phenol and a hydrogen atom is substituted for a bromine atom in the 2 and 6 positions
  - There is a group already on the 4th carbon atom in the ring so the bromine atom does not substitute here
  - The -OH group is 2,4,6 directing
  - The ketone group is also 3,5 directing so this corresponds to where the bromine atoms have been substituted
- I_2 / NaOH (aq) are the reagents involved in the iodoform reaction, the ketone

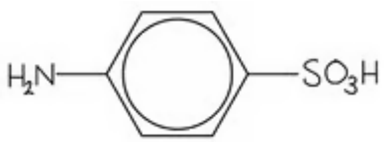
group in the compound will react with these reagents
- The reaction involves a halogenation and hydrolysis step
  - In the halogenation step, all three H-atoms in the R group are replaced with iodine atoms, forming a $-CI_3$ group
  - The intermediate compound is hydrolysed by an alkaline solution to form a sodium salt ($RCO_2^- Na^+$) and a yellow precipitate of CHI_3

5c

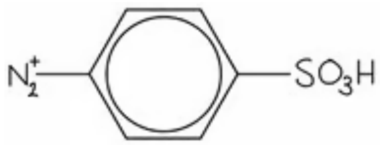
c)

i) The structures are:

Compound **J**:

-  ; [1 mark]

Compound **K**:

-  ; [1 mark]

ii) The reagents and conditions required are:

Step 1

- $NaNO_2$ + HCl
  OR
  HNO_2; [1 mark]
- T < 10 °C; [1 mark]

Step 2

- (Add K to a solution of **G**) in NaOH (aq); [1 mark]

[Total: 5 marks]

- You are told that compound **J** is an amine and is converted to compound **K** which then forms an azo dye

- In order to form an azo dye, the amine group must be converted to a diazonium ion
- This then can bond with compound **G** to form the azo dye **H**
  - This must take place in alkaline conditions, hence the requirement for NaOH (aq)
- **Remember:** HNO_2 must be formed 'in situ' in the reaction vessel as it is very unstable
  - So, $NaNO_2$ and HCl must be used to form the acid
  - This also must take place at a low temperature, below 10 °C

5d

d) Dye **H** is very stable because:

- Delocalised electrons in the π bonding systems of the two benzene rings; [1 mark]
- Extended through the -N=N- which acts as a bridge between the two rings; [1 mark]

[Total: 2 marks]

- Azo dyes contain benzene rings